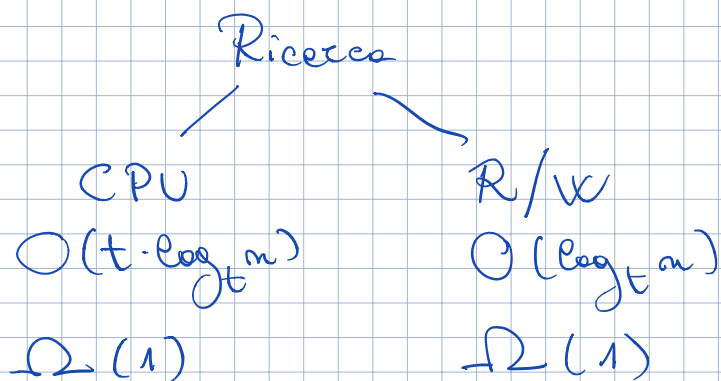
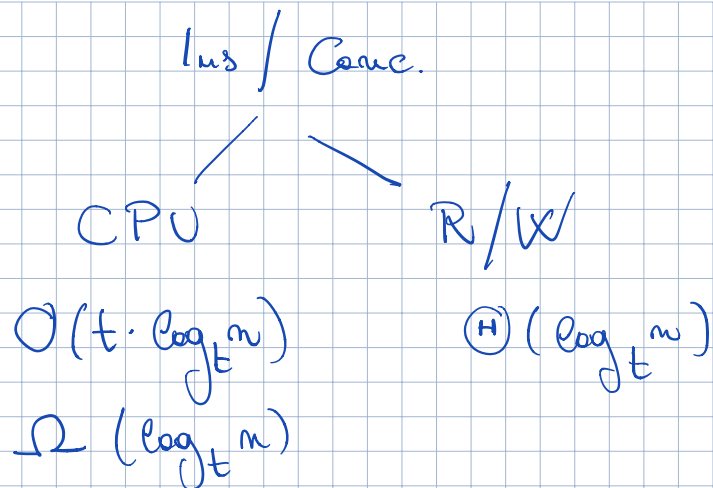


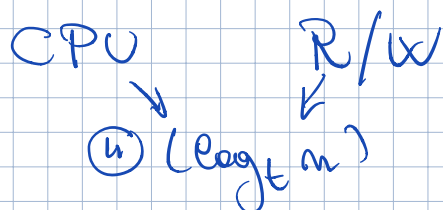


~ . ~ . ~

Es.

Dato T B-tree di grado t
determinare l'altezza di T

Scendo a xx (tutti i rami hanno
la stessa lung.)



RBT e BST $\sim \Theta(n)$

RBT altezza media $\sim \Theta(\log n)$

```
Btree Height (T, t) {  
    x ← DiskRead (T.root)    % x ← T.root  
    h ← 0  
    while (not x.leaf) {  
        x ← DiskRead (x.ci)  
        h ← h + 1  
    }  
    return h  
}
```



$h = 0$



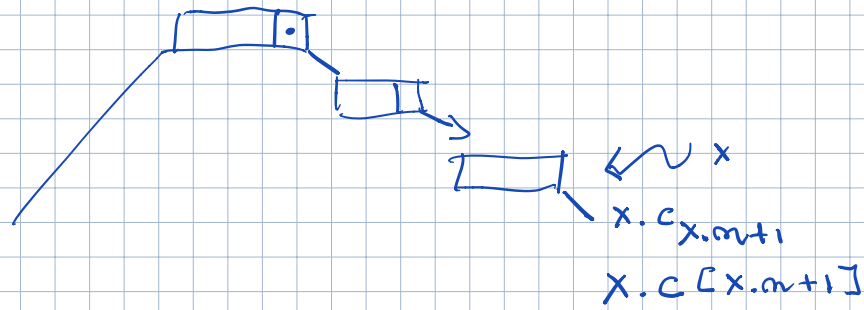
$h = 1$

$\sim \sim$

Es.

Dato T B-tree di grado t

- cercare la chiave massima in T
minimo



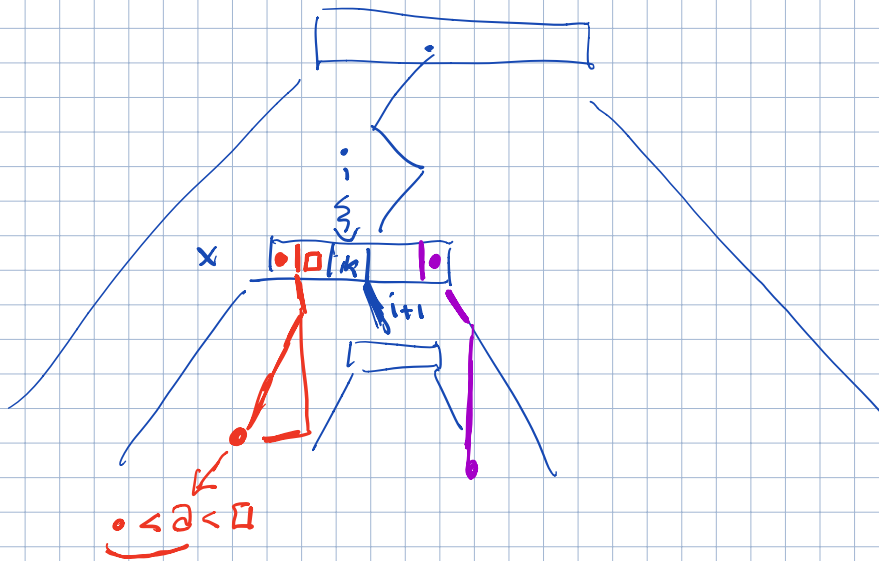
CPU $\downarrow$
 $\textcircled{u} (\log_t m)$

$R/w \downarrow$

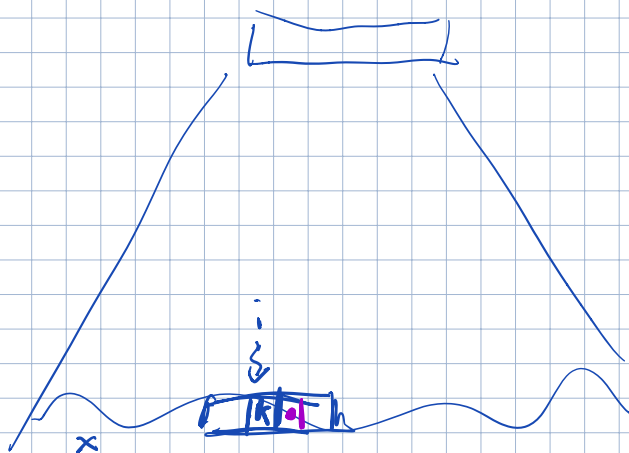
Ex.

Dato T B-tree di grado t

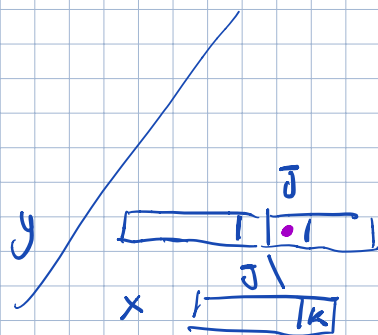
Dato k chiave di T cercare k'
 successore di T



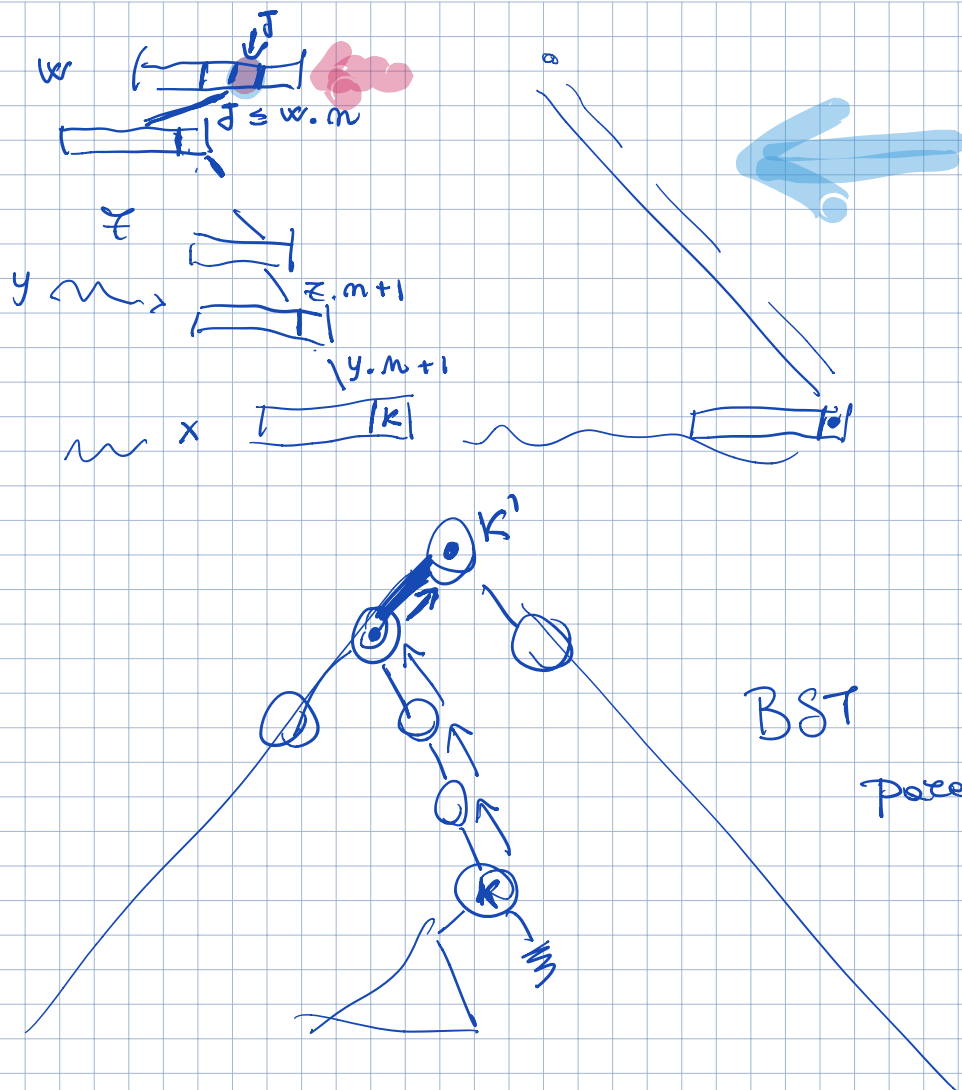
- x non foglia
 - scendo in $x.c_{i+1}$ e cerco il min



- x foglia
 - $i < x.m \rightarrow x.key_{i+1}$
 - $i = x.m$



x foglia
 k ultima
 chiave di x
 x figlio J -esimo
 di y
 $J \leq y.m$



Btree Successor (T, t, k) $\{$ $\% k \in \text{in } T$
 $x \leftarrow \text{DiskRead}(T.\text{root})$
 $k' \leftarrow +\infty$
 • while (not $x.\text{leaf}$) $\{$
 $i \leftarrow 1$
 $\text{while } (i \leq x.m \ \&\& \ x.\text{key}_i \leq k) \{$
 $i \leftarrow i+1$
 $\}$

```

    if ( $i \leq x.m \wedge x.key_i = k$ ) {
         $y \leftarrow \text{DiskRead}(x.c_i)$ 
        return BtreeMin( $y, t$ )
    }

    if ( $i \leq x.m$ ) {
         $k' \leftarrow x.key_i$ 
    }

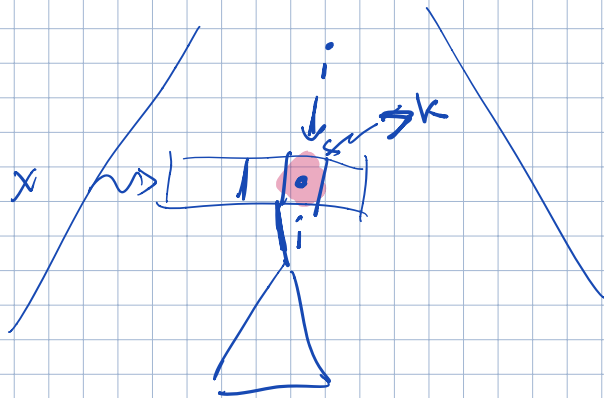
     $x \leftarrow \text{DiskRead}(x.c_i)$ 
}

i ← 1
while ( $i \leq x.m \wedge x.key_i \neq k$ ) {
    i ← i + 1
}

if ( $i \leq x.m$ ) {
    return  $x.key_{i+1}$ 
} else {
    return  $k'$ 
}
}

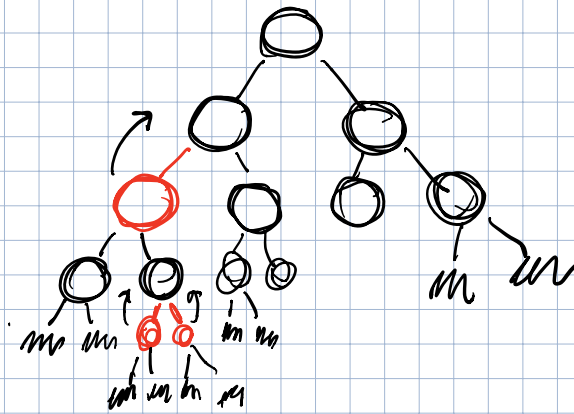
```

CPU $\Downarrow$ $O(t \log_t m) \Omega(\log_t m)$
 R/W $\Theta(\log_t m)$

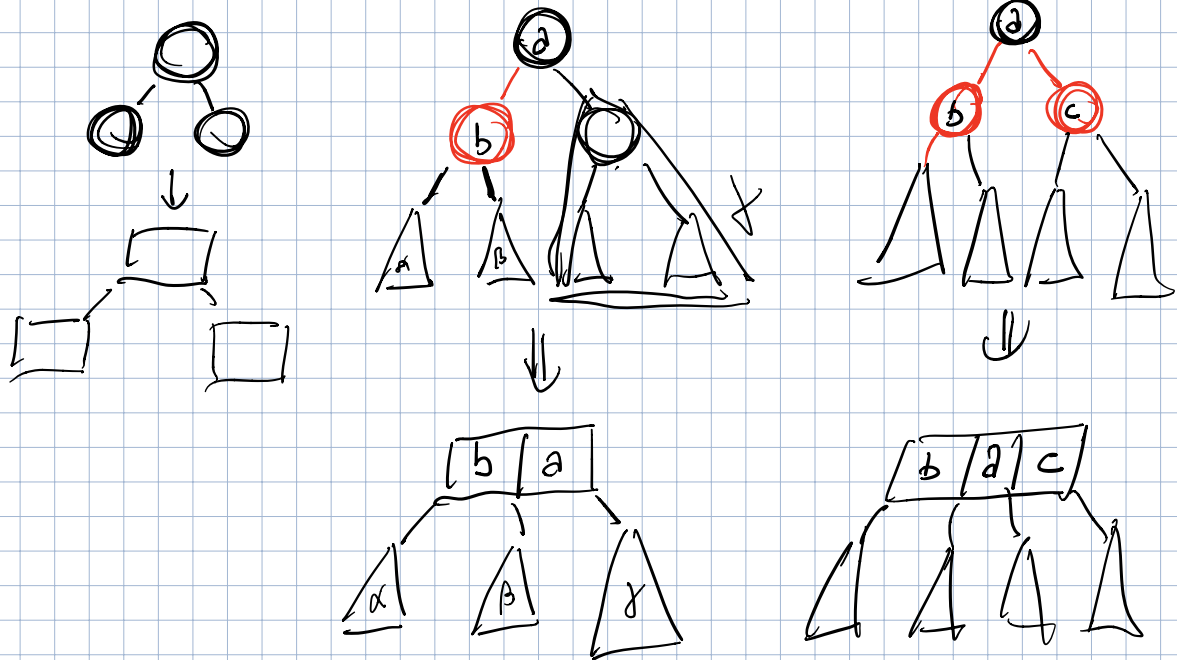


~ , ~ , ~ ,

Che legame c'è tra B-tree e
RBT ?



B-tree di grado 2



Ottengo un B-tree di grado 2
e altezza pari all'altezza nera del RBT

CPU

R/W

$\Theta(n)$

È vero ??

Un Btree di grado 2 si può trasformare
in un RBT?

